

Market risk Project

- A. Calculation of the Value at Risk for $\alpha=5\%$ with KDE. First, we calculated the log returns $(\ln(P_t/P_{t-1}))$. Then, we calculate the Bandwidth h . We initialize x at $\text{Min}(\text{returns}) - 5h$ with a step of 0.0003 to be able to cover the entire CDF. We apply the CDF F to all x with $F(x) = \frac{1}{n} \sum_{i=1}^n G(hx - r_i)$, where G is the CDF of $F(x) \leq 0.05$, this corresponds to $F^{-1}(\alpha)$. By definition, for a gain with α close to 0, $\text{Var}(X) = -F^{-1}(\alpha)$. We find $\text{VaR}(X) \approx 3.89\%$.

Furthermore, we verify the number of times the return is strictly lower than Q_α (equivalent to $-r_t > \text{VaR}$). We find a violation rate of $1.57\% < 5\%$; the model therefore overestimates the risk.

- B. Calculation of the Expected Shortfall. We use the formula with the expectation, which is: $ES = E[-r \mid r \leq -\text{Var}(X)] = -E[r \mid r \leq Q_\alpha]$. We notice that ES is approximately 5.45%, which is significantly higher than the $\text{VaR}(X)$.

C.

We sort the returns in ascending order; we isolate the losses ($r < 0$) by applying the absolute value and the gains ($r \geq 0$). In both cases, we take $k = \sqrt{n}$ with n being the number of observations. We calculate the estimator using the formula:

$$\hat{\xi} = \frac{1}{\ln 2} \ln \left(\frac{X_{n-k+1} - X_{n-2k+1}}{X_{n-2k+1} - X_{n-4k+1}} \right)$$

We find approximately 0.41 for the losses and 0.71 for the gains. Both are strictly greater than 0, therefore the nature of the extreme gains and losses belongs to the Fréchet domain of attraction. We use this formula to calculate the VaRs as a function of the different values of p : $\text{VaR}(p) =$

$$\frac{\left(\frac{k}{n(1-p)}\right)^{\xi_P - 1}}{1 - 2^{-\xi_P}} (A - B) + A \text{ où } A = X_{n-k+1:n}, B = X_{n-2k+1:n}.$$

- D. With the solver, we determined the values of $G_0=0$ and $\beta = 0.5$. $L=50$ and $\gamma = 0.5$ are given in the problem statement.
- E. For each series, we calculated the average price and the log return of these average prices. We then apply the Haar wavelet transform to the return series, and for each time scale, we calculate the associated wavelet coefficients. Then, we estimate the correlation between each pair of exchange rates. We observe an increasing correlation with the scale; this is the Epps effect. This effect can be explained by transaction desynchronization phenomena.

The Hurst exponents are:

- GBPEUR : $H \approx 0,515$
- SEKEUR : $H \approx 0,404$
- CADEUR : $H \approx 0,401$

GBPEUR is close to a random walk, whereas SEK and CADEUR are anti-persistent.

The annualized volatilities (standard) are:

$$\sigma_{an} = \sigma_{jour} \times \sqrt{252}$$

- GBPEUR : $\approx 12.67\%$
- SEKEUR : $\approx 4.65\%$
- CADEUR : $\approx 8.32\%$

The annualized volatilities (Hurst) are: $\sigma_{ann,iid} = \sigma_{day} \times \sqrt{252}$

- GBPEUR : $\approx 13.80\%$
- SEKEUR : $\approx 2.74\%$
- CADEUR : $\approx 4.81\%$